

Python Dictionary Practice — Basic to Hard

50 Python Dictionary Practice Questions

Arranged from Basic → Hard • Each question includes example input and expected output.

Level	Difficulty	Focus	Questions
1	Basic	Dictionary creation, access, update, insertion, deletion	10
2	Basic → Intermediate	Counting, searching, iteration, merging and filtering	10
3	Intermediate	Frequency maps, sorting by values, nested dictionaries and transformations	10
4	Advanced	Aggregation, grouping, nested structures and multi-step processing	8
5	Hard	Complex dictionary algorithms and optimization-style problems	12

Level 1 — Basic

Dictionary creation, access, update, insertion, deletion.

Q No.	Question	Example Input	Example Output
1	Create a dictionary with three key-value pairs.	Input: name='Asha', age=20, city='Delhi'	Output: {'name':'Asha','age':20, 'city':'Delhi'}
2	Access a value using its key.	Input: D={'a':10,'b':20}, key='b'	Output: 20
3	Add a new key-value pair.	Input: D={'a':1}, D['b']=2	Output: {'a':1,'b':2}
4	Update the value of an existing key.	Input: D={'score':50}, score=80	Output: {'score':80}
5	Check whether a key exists.	Input: D={'a':1,'b':2}, key='c'	Output: False
6	Delete a key from a dictionary.	Input: D={'a':1,'b':2}, delete='a'	Output: {'b':2}
7	Find the number of key-value pairs.	Input: D={'a':1,'b':2,'c':3}	Output: 3
8	Get all keys.	Input: D={'name':'Raj','age':21}	Output: ['name','age']
9	Get all values.	Input: D={'a':10,'b':20}	Output: [10,20]
10	Clear a dictionary.	Input: D={'a':1,'b':2}	Output: {}

Level 2 — Basic → Intermediate

Counting, searching, iteration, merging and filtering.

Q No.	Question	Example Input	Example Output
1	Count character frequencies in a string.	Input: 'aabbc'	Output: {'a':2,'b':2,'c':1}
2	Count word frequencies in a sentence.	Input: 'cat dog cat bird'	Output: {'cat':2,'dog':1,'bird':1}
3	Find the key with the maximum value.	Input: D={'a':10,'b':25,'c':15}	Output: 'b'
4	Find the key with the minimum value.	Input: D={'a':10,'b':25,'c':15}	Output: 'a'
5	Merge two dictionaries.	Input: A={'a':1}, B={'b':2}	Output: {'a':1,'b':2}
6	Filter dictionary items whose values are greater than 50.	Input: {'a':40,'b':60,'c':80}	Output: {'b':60,'c':80}
7	Create a dictionary from two lists.	Input: Keys=['a','b'], Values=[10,20]	Output: {'a':10,'b':20}
8	Invert a dictionary with unique values.	Input: {'a':1,'b':2}	Output: {1:'a',2:'b'}
9	Find common keys of two dictionaries.	Input: A={'a':1,'b':2}, B={'b':3,'c':4}	Output: {'b'}
10	Sum all numeric dictionary values.	Input: {'a':10,'b':20,'c':5}	Output: 35

Level 3 — Intermediate

Frequency maps, sorting by values, nested dictionaries and transformations.

Q No.	Question	Example Input	Example Output
1	Find the second highest value and its key.	Input: {'a':10,'b':30,'c':20}	Output: ('c',20)
2	Sort a dictionary by values ascending.	Input: {'a':3,'b':1,'c':2}	Output: {'b':1,'c':2,'a':3}
3	Sort a dictionary by keys descending.	Input: {'b':2,'a':1,'c':3}	Output: {'c':3,'b':2,'a':1}
4	Combine dictionaries by adding values for common keys.	Input: A={'a':10,'b':20}, B={'b':5,'c':7}	Output: {'a':10,'b':25,'c':7}
5	Find keys whose values are duplicated.	Input: {'a':1,'b':2,'c':1,'d':3}	Output: {'a','c'}
6	Group words by their first letter.	Input: ['apple','ant','ball','bat']	Output: { 'a':['apple','ant'], 'b':['ball','bat']}
7	Create a frequency dictionary for vowels.	Input: 'programming'	Output: {'a':1,'i':1,'o':1}
8	Find all keys whose values are above the average.	Input: {'a':10,'b':20,'c':30}	Output: {'c'}
9	Remove dictionary entries with zero values.	Input: {'a':0,'b':2,'c':0}	Output: {'b':2}
10	Find the most frequent value in a dictionary.	Input: {'a':1,'b':2,'c':1,'d':1}	Output: 1

Level 4 — Advanced

Aggregation, grouping, nested structures and multi-step processing.

Q No.	Question	Example Input	Example Output
1	Invert a dictionary where multiple keys can share the same value.	Input: {'a':1,'b':2,'c':1}	Output: {1:['a','c'],2:['b']}
2	Merge a list of dictionaries by summing common keys.	Input: [{'a':1,'b':2},{'b':3,'c':4}]	Output: {'a':1,'b':5,'c':4}
3	Find the top k keys by value.	Input: {'a':5,'b':9,'c':7,'d':3}, k=2	Output: ['b','c']
4	Create nested dictionary counts from a list of (category,item) pairs.	Input: [('fruit','apple'),('fruit','banana'),('fruit','apple')]	Output: {'fruit':{'apple':2,'banana':1}}
5	Find the intersection of keys across many dictionaries.	Input: [{'a':1,'b':2},{'b':3,'c':4},{'b':5,'d':6}]	Output: {'b'}
6	Flatten a one-level nested dictionary.	Input: {'user':{'name':'Sam','age':20}}	Output: {'user.name':'Sam','user.age':20}
7	Find the key whose value list has the largest sum.	Input: {'a':[1,2],'b':[5,1],'c':[2,2]}	Output: 'b'
8	Group students by grade using a dictionary.	Input: [('A',90),('B',75),('C',90)]	Output: {90:['A','C'],75:['B']}

Level 5 — Hard

Complex dictionary algorithms and optimization-style problems.

Q No.	Question	Example Input	Example Output
1	Find the longest consecutive sequence of dictionary keys that are integers.	Input: {1:'a',2:'b',3:'c',5:'e'}	Output: 3
2	Find a pair of keys whose values sum to a target.	Input: {'a':2,'b':7,'c':4,'d':5}, target=9	Output: ('a','b')
3	Build a word-to-index dictionary from a sentence.	Input: 'red blue red green'	Output: {'red':[0,2],'blue':[1],'green':[3]}
4	Find the smallest set of dictionary keys whose values reach a target sum.	Input: {'a':4,'b':7,'c':9}, target=13	Output: {'a','c'}
5	Compare two nested dictionaries and return changed keys.	Input: A={'x':{'a':1}}, B={'x':{'a':2}}	Output: ['x.a']
6	Find the first key whose value reaches or exceeds a target while scanning insertion order.	Input: {'a':2,'b':4,'c':7}, target=6	Output: 'c'
7	Merge nested dictionaries recursively.	Input: A={'x':{'a':1}}, B={'x':{'b':2}}	Output: {'x':{'a':1,'b':2}}
8	Find keys with the same value in two dictionaries.	Input: A={'a':1,'b':2}, B={'x':2,'y':3}	Output: [['b','x']]
9	Create a dictionary mapping each value to all positions where it occurs in a list.	Input: ['a','b','a','c','b']	Output: {'a':[0,2],'b':[1,4],'c':[3]}
10	Find the dictionary key with the highest ratio of value to key length.	Input: {'aa':10,'b':4,'cccc':15}	Output: 'b'
11	Find all keys whose values are prime numbers.	Input: {'a':7,'b':8,'c':11,'d':12}	Output: {'a','c'}
12	Find the maximum depth of a nested dictionary.	Input: {'a':{'b':{'c':1}}}	Output: 3